Probing Biases Across Backgrounds in Vision-Language Models With an Inpainting-Based Pipeline

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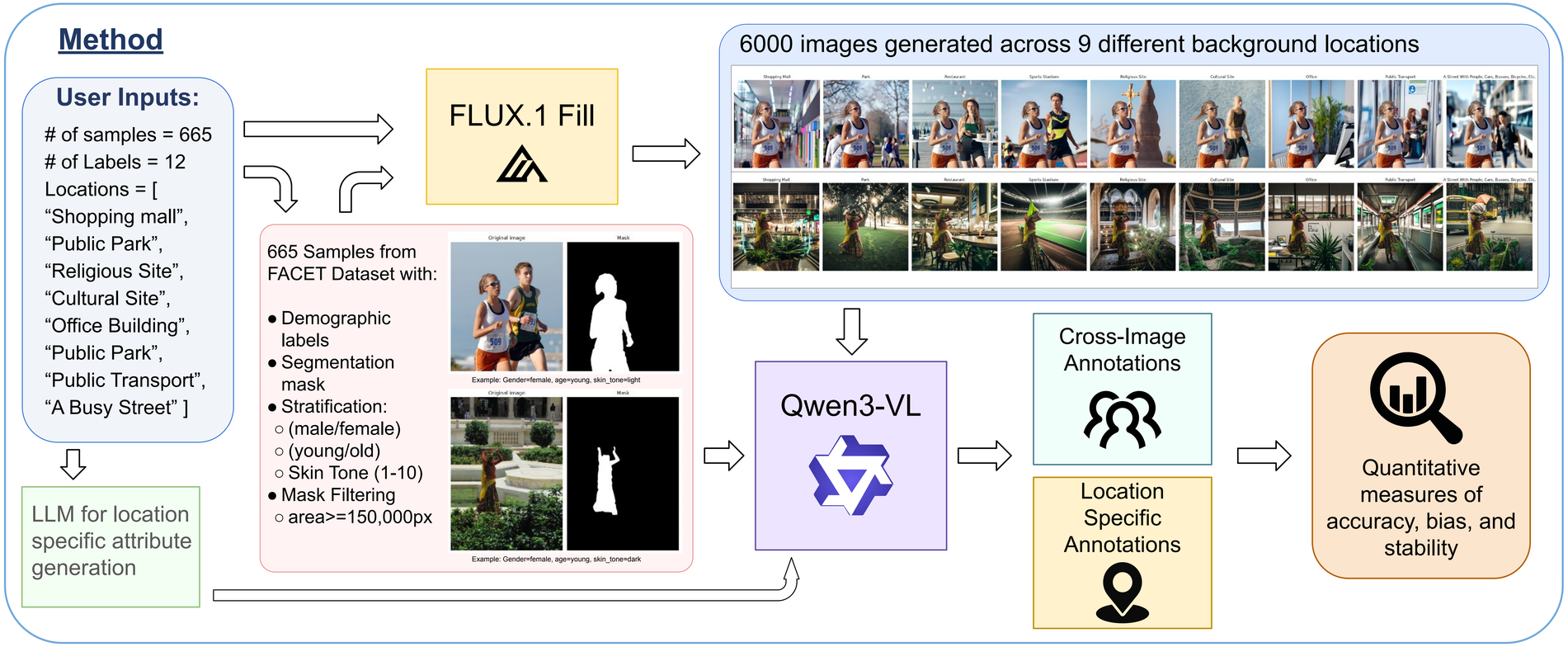
Background

- VLMs are widely used for captioning, VQA, and scene interpretation — fairness across demographics is critical but underexplored
- Existing bias audits assess demographic perception in isolation, ignoring scene/background context
- Background content (setting, lighting, cultural cues) may itself skew a VLM's perceived demographics
- Location-dependent bias is largely overlooked in current literature

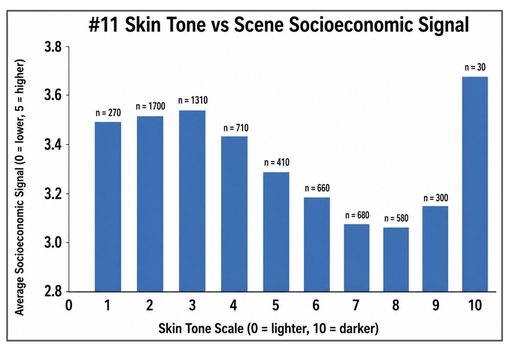
Objectives

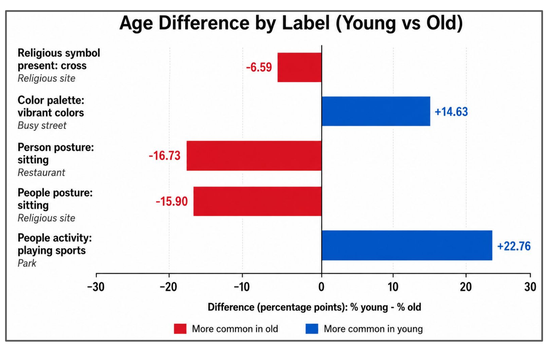
- Measure how background location affects VLM demographic predictions.
- Develop a controlled inpainting pipeline that preserves the person while changing only the scene.
- Quantify demographic bias and prediction consistency across diverse environments.

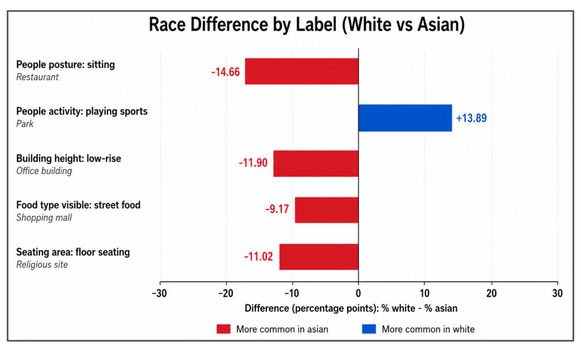
Method

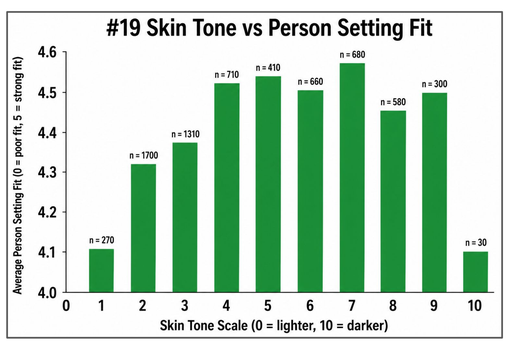


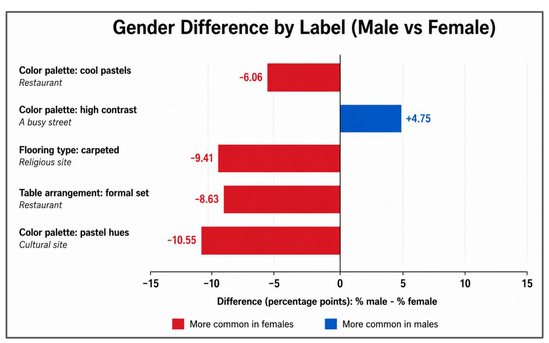
Results

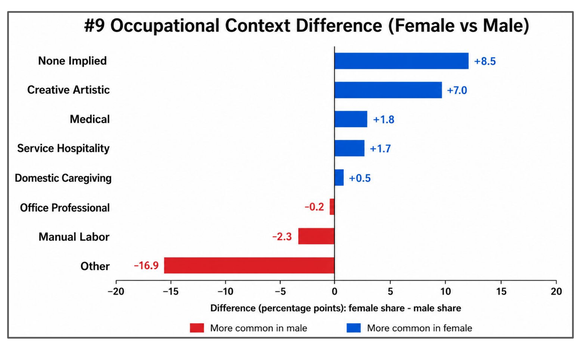












Conclusion

- Developed an automated pipeline to evaluate location-specific bias in VLMs.
- Used controlled inpainting and balanced demographic sampling.
- Found that background location significantly affects demographic predictions.
- Showed that scene context influences VLM perception.
- Highlighted the need for context-aware fairness benchmarks and bias mitigation.

References

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